LEAKLISTEN · USER REQUIREMENTS

LeakListen user requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-01	Amish Chadha	First user requirements, written in the kit 1.9.1 upgrade from the users and stakeholders in LKL-PRB-001 and the TRL 2 and TRL 3 requirements R1 to R15; used to score the concept options (LKL-OPT-001)

LeakListen user requirements

What users need, written from their side. The sixteen needs come from the users and stakeholders in the problem statement (LKL-PRB-001, Table 1 and Constraints) and from what the earlier requirements R1 to R15 already assumed, and they are used to score the concept options (LKL-OPT-001). Each one has at least one product requirement in LKL-PRS-001 (links in LKL-RTM-001). They are a desk draft: no utility has reviewed them, and the first candidate partner, a UK water company with metallic mains and district metered areas, has not been approached (LKL-DEC-001, 2026-10-02). Method: STANDARDS section 22.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Performance	Hears leaks before they surface	The logger tells the leakage team that a hidden leak is near it, at night, without anyone listening on site	High	Small and mid-sized utility; leak detection technician (PRB Table 1)
UR-002	Performance	Works on plastic mains too	Plastic (PVC and PE) mains are covered as well as metal ones, at least through a variant	Medium	Small and mid-sized utility; PRB operating context; LKL-DDR-002
UR-003	Deployment and assembly	Placed from the surface	Nobody enters a chamber or a trench to place, service or recover a logger	High	Leak detection technician; PRB Constraints and Safety
UR-004	Deployment and assembly	Quick to place	One crew places or recovers a logger in minutes, with a pole or valve key and no pipe work	Medium	Leak detection technician
UR-005	Performance	Reports by itself	A short nightly summary reaches the utility's server over a low-power radio, with no visit and no cabling	High	Utility manager; PRB Constraints
UR-006	Environment and durability	Survives the chamber	It works in a dark, damp chamber that may flood, from cold winters to warm summers	High	Leak detection technician; PRB operating context
UR-007	Maintenance	Runs for years unattended	One battery lasts years, because there is no power or sun in a chamber	High	Utility manager; PRB Constraints
UR-008	Cost	Affordable for a small utility	A fleet of loggers costs far less than commercial noise loggers and their subscriptions	High	Small and mid-sized utility; community scheme (PRB problem)
UR-009	Regulatory	Raw sound stays on the device	Nothing that could be heard as speech leaves the logger	High	Utility manager; regulator; LKL-DDR-001 D6
UR-010	Usability	Open data and standard radio	Works with any LoRaWAN network server and gives researchers open, documented data	Medium	Researchers and students; community scheme
UR-011	Safety	Safe to handle	The lithium cell, the strong magnet and the lifting of covers do not hurt the crew	High	Leak detection technician; PRB Safety
UR-012	Safety	Does not touch the water	Nothing touches drinking water or is fixed to the pipe, so water quality and the asset are untouched	High	Utility manager; PRB Constraints
UR-013	Usability	Few false alarms	A flag means a crew should go and look; the flag list is short and credible	Medium	Utility manager; leak detection technician
UR-014	Packaging and transport	Small and light	One person carries the logger set and it fits through a valve box or chamber neck	Medium	Leak detection technician
UR-015	Maintenance			Medium	

ID	Category	Title	Need	Priority	Source
		Open and repairable	A utility or a workshop can inspect, repair and adapt the logger, unlike closed commercial loggers		Community scheme; researchers; PRB problem
UR-016	Regulatory	Legal to place and to transmit	It uses licence-exempt radio within the regional power limit and is placed only with the utility's permission	Medium	Regulator; utility manager; PRB Constraints

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.